

Acc501 Final Term

Lecture 21

Different Types of Bonds

Government Bonds

- **Treasury Notes and Bonds:** Used by the government to borrow money for more than one year.
- **Treasury Issues:** Have no default risk.
- **Tax Exemptions:** Often exempt from income taxes.

Zero Coupon Bonds

- **No Coupons:** Do not pay periodic interest.
- **Discounted Price:** Sold at a lower price than face value.
- **Example:** A \$1000 face value, 5-year bond sold for \$497, yielding 15% to maturity.
- **Interest Calculation:** The difference between face value and purchase price is the total interest paid.
- **Tax Considerations:** Interest is deducted yearly for tax purposes, although no actual interest is paid, resulting in lower yields.

Zero Coupon Bonds Example

Year	Beginning Value	Ending Value	Implicit Interest Expense	Straight Line Interest Expense
1	\$497	\$572	\$75	\$100.60
2	\$572	\$658	\$86	\$100.60
3	\$658	\$756	\$98	\$100.60
4	\$756	\$870	\$114	\$100.60
5	\$870	\$1000	\$130	\$100.60
Total	\$503	\$503		

Floating Rate Bonds

- **Adjustable Coupons:** Coupon payments adjust based on an interest rate index, such as treasury bills.
- **Put Provision:** Holders can redeem the bond at par on the coupon payment date after a specified period.
- **Collar:** Coupon rate is capped with minimum (floor) and maximum (ceiling) limits.
- **Inflation-Linked Bonds:** Coupons adjust according to the rate of inflation, with possible principal adjustments.

Other Types of Bonds

- **Income Bonds:** Coupon payments depend on the issuer's income.
- **Convertible Bonds:** Can be swapped for a fixed number of shares before maturity at the holder's option.
- **Put Bonds:** Allow the holder to force the issuer to buy back the bond at a stated price.

Inflation and Interest Rates

Real vs. Nominal Rates

- **Real Rates:** Adjusted for inflation.
- **Nominal Rates:** Not adjusted for inflation.

Example:

- **Inflation Rate:** 5%
- **Investment:** \$100 today will be worth \$115.50 in one year (nominal return of 15.5%).
- **Real Rate of Return:** Considering inflation, the real rate of return is 10%.

Pen Example:

- **Initial Pen Cost:** \$5 per pen (20 pens for \$100).
- **End of Year Pen Cost:** \$5.25 per pen.
- **Investment:** \$115.50 buys 22 pens, up from 20, giving a real return of 10%.

The Fisher Effect

- **Relationship:** Describes the relationship between real and nominal returns.
- **Formula:** $1+R=(1+r)\times(1+h)$
 - **R:** Nominal return
 - **r:** Real return
 - **h:** Inflation rate

Example Calculation:

- **Nominal Rate (R):** 15.5%
- **Inflation Rate (h):** 5%
- **Real Rate (r):**

$$1+0.155=(1+r)\times(1+0.05)$$

$$=10\%$$

- **Approximation:**

$$R \approx r + h$$

MCQs (Multiple Choice Questions)

Here are some sample multiple choice questions (MCQs) based on the provided text:

1. **What type of bond does not pay periodic interest and is sold at a significant discount to its face value?**
 - A. Government Bonds
 - B. Zero Coupon Bonds
 - C. Floating Rate Bonds
 - D. Convertible Bonds

Answer: B. Zero Coupon Bonds

2. **Which bond allows the holder to force the issuer to buy back the bond at a stated price?**
 - A. Convertible Bonds
 - B. Put Bonds
 - C. Income Bonds
 - D. Floating Rate Bonds

Answer: B. Put Bonds

3. **Which type of bond has an adjustable coupon rate based on an interest rate index?**
 - A. Zero Coupon Bonds
 - B. Government Bonds
 - C. Floating Rate Bonds
 - D. Income Bonds

Answer: C. Floating Rate Bonds

4. **Which of the following statements is true about government bonds?**
 - A. They have high default risk.
 - B. They are typically taxable.
 - C. They are sold by corporations.
 - D. They are often exempt from income taxes.

Answer: D. They are often exempt from income taxes.

5. **What is the term for the interest rate or rate of return that has been adjusted for inflation?**
 - A. Nominal Rate
 - B. Real Rate
 - C. Coupon Rate
 - D. Implicit Rate

Answer: B. Real Rate

Additional multiple-choice questions based on the provided text:

6. **What feature of floating rate bonds allows the holder to redeem the note at par on the coupon payment date after a specified period?**

- A. Convertible Provision
- B. Put Provision
- C. Call Provision
- D. Inflation Adjustment

Answer: B. Put Provision

7. **Which of the following best describes the Fisher Effect?**

- A. It explains the difference between real and nominal rates of return.
- B. It is the interest rate paid on zero coupon bonds.
- C. It determines the coupon rate for floating rate bonds.
- D. It is the method of calculating the straight-line interest expense.

Answer: A. It explains the difference between real and nominal rates of return.

8. **What happens to the principal value of an inflation-linked bond?**

- A. It remains constant over time.
- B. It decreases with inflation.
- C. It is adjusted according to the rate of inflation.
- D. It is adjusted according to the nominal interest rate.

Answer: C. It is adjusted according to the rate of inflation.

9. **Which type of bond can be swapped for a fixed number of shares before maturity at the holder's option?**

- A. Put Bonds
- B. Income Bonds
- C. Convertible Bonds
- D. Zero Coupon Bonds

Answer: C. Convertible Bonds

10. **If the nominal return on an investment is 15.5% and the inflation rate is 5%, what is the real rate of return?**

- A. 15.5%
- B. 5%
- C. 10%
- D. 20.5%

Answer: C. 10%

11. **Which of the following is a characteristic of income bonds?**

- A. They can be converted into shares.
- B. They do not pay periodic interest.

- C. Their coupon payments depend on the issuer's income.
- D. They adjust their coupon rate based on an interest rate index.

Answer: C. Their coupon payments depend on the issuer's income.

12. What is the main advantage of government bonds?

- A. High returns
- B. Tax exemptions
- C. High default risk
- D. Variable interest rates

Answer: B. Tax exemptions

13. Which bond type is issued at a discount and does not make periodic interest payments?

- A. Convertible Bonds
- B. Income Bonds
- C. Zero Coupon Bonds
- D. Floating Rate Bonds

Answer: C. Zero Coupon Bonds

14. What is the main characteristic of a nominal interest rate?

- A. It is adjusted for inflation.
- B. It does not account for inflation.
- C. It is the real return rate.
- D. It is lower than the real rate.

Answer: B. It does not account for inflation.

15. Which component is not part of the nominal rate according to the Fisher Effect?

- A. Real rate on investment
- B. Compensation for inflation on the principal
- C. Compensation for inflation on income earned
- D. Adjusted coupon rate

Answer: D. Adjusted coupon rate

Lecture 22

Term Structure of Interest Rates

Term Structure of Interest Rates

- **Definition:** The relationship between short- and long-term interest rates.
- **Nominal Interest Rates:** Based on default-free, pure discount bonds of all maturities.
- **Pure Interest Rates:** Involve no risk of default and a single, lump-sum future payment.

- **Upward Sloping:** Long-term rates higher than short-term rates.
- **Downward Sloping:** Long-term rates lower than short-term rates.

Determinants of Term Structure

1. **Real Rate of Interest:**
 - High real rates lead to higher overall interest rates and vice versa.
 - Influences the overall level of interest rates, not the shape of the term structure.
2. **Expected Inflation:**
 - Strongly influences the shape of the term structure.
 - Higher expected inflation leads to higher long-term rates, creating an upward-sloping term structure.
3. **Interest Rate Risk:**
 - Long-term bonds have greater risk of loss due to changes in interest rates than short-term bonds.
 - Investors demand extra compensation (interest rate risk premium) for bearing this risk.
 - The longer the term to maturity, the greater the interest rate risk and premium.
 - Interest rate risk premium increases at a decreasing rate.

Bond Yields and the Yield Curve

- **Treasury Yield Curve:** Plots treasury yields relative to maturity.
- **Term Structure vs. Yield Curve:**
 - Term structure is based on pure discount bonds.
 - Yield curve is based on coupon bond yields.
- **Treasury Notes and Bonds Features:**
 - Default-free, taxable, highly liquid.
 - Default risk premium: Extra yield demanded by investors as compensation for default risk.
 - Taxability premium: Extra yield demanded on taxable bonds as compensation for unfavorable tax treatment.
 - Liquidity premium: Compensation demanded for bonds with varying degrees of liquidity.

Common Stock Valuation

Valuation Challenges

- Promised cash flows are not known in advance.
- Common stock has no maturity.
- Market rate of return is not easily observed.

Cash Flows Example

- **Stock Purchase:** Plan to sell in a year with an expected dividend and ending value.
- **Example:** Stock worth \$70 at the end of the year with a \$10 dividend.

- **Required Return:** If a 25% return is required, the maximum price to pay for the stock today is \$64.

General Valuation Formula

- $$P_0 = \frac{D_1 + P_1}{1 + R}$$
 - P0: Current price of stock
 - P1: Price in one period
 - D1: Dividend paid at the end of the period
 - RR: Market rate of return
- Extending this to multiple periods:

$$P_0 = \frac{D_1}{1 + R} + \frac{D_2}{1 + R} + \frac{D_3}{1 + R} \dots$$

Zero Growth Stocks

- **Definition:** Stocks with a constant dividend.

$$P_0 = \frac{D_1}{R}$$

- **Value Formula:**
 - D: Dividend per period
 - R: Required rate of return
- **Example:** CVP Corporation pays a \$10 dividend indefinitely. If the required return is 20%, the stock value is \$50 per share.

Multiple Choice Questions

1. **What does the term structure of interest rates represent?**
 - A. The relationship between different bond issuers
 - B. The relationship between short- and long-term interest rates
 - C. The risk of default on bonds
 - D. The yield on government bonds

Answer: B. The relationship between short- and long-term interest rates

2. **What is an upward-sloping term structure indicative of?**
 - A. Higher short-term rates than long-term rates
 - B. Higher long-term rates than short-term rates
 - C. Equal short- and long-term rates
 - D. Decreasing interest rates

Answer: B. Higher long-term rates than short-term rates

3. **Which of the following does not influence the shape of the term structure?**

- A. Real rate of interest
- B. Expected inflation
- C. Interest rate risk
- D. Default risk

Answer: D. Default risk

4. **What is the interest rate risk premium?**

- A. Compensation for default risk
- B. Compensation for unfavorable tax treatment
- C. Extra compensation for bearing the risk of interest rate changes
- D. The difference between real and nominal rates

Answer: C. Extra compensation for bearing the risk of interest rate changes

5. **What is the shape of the yield curve a reflection of?**

- A. The term structure of interest rates
- B. Default risk premium
- C. Taxability premium
- D. Liquidity premium

Answer: A. The term structure of interest rates

6. **What does a default risk premium compensate for?**

- A. The possibility of tax changes
- B. The risk of a bond issuer defaulting
- C. The risk of interest rate changes
- D. The difference in bond liquidity

Answer: B. The risk of a bond issuer defaulting

7. **Which feature is common to treasury notes and bonds?**

- A. High default risk
- B. High liquidity
- C. Exemption from taxes
- D. Variable interest rates

Answer: B. High liquidity

8. **What type of premium is demanded for bonds with unfavorable tax treatment?**

- A. Default risk premium
- B. Liquidity premium
- C. Taxability premium
- D. Interest rate risk premium

Answer: C. Taxability premium

9. **Which of the following is not a reason why common stock valuation is difficult?**

- A. Promised cash flows are known in advance
- B. Common stock has no maturity
- C. Market rate of return is not easily observed
- D. Cash flows are uncertain

Answer: A. Promised cash flows are known in advance

10. **How is the value of zero growth stock calculated?**

- A. As an ordinary perpetuity
- B. Based on future dividends only
- C. Using the Fisher Effect
- D. By discounting future stock prices

Answer: A. As an ordinary perpetuity

10 More multiple-choice questions based on the provided text:

Term Structure of Interest Rates and Bond Valuation

1. **What primarily influences the overall level of interest rates?**

- A. Expected inflation
- B. Real rate of interest
- C. Interest rate risk
- D. Liquidity premium

Answer: B. Real rate of interest

2. **What do investors demand to compensate for the erosion of dollar returns due to inflation?**

- A. Real rate premium
- B. Liquidity premium
- C. Default risk premium
- D. Inflation premium

Answer: D. Inflation premium

3. **What happens to the interest rate risk premium as the term to maturity increases?**

- A. It decreases at a decreasing rate.
- B. It increases at an increasing rate.
- C. It increases at a decreasing rate.
- D. It remains constant.

Answer: C. It increases at a decreasing rate.

4. **Which of the following is true about the yield curve?**

- A. It is always upward sloping.
- B. It is based on coupon bond yields.
- C. It only applies to zero-coupon bonds.
- D. It is not influenced by inflation expectations.

Answer: B. It is based on coupon bond yields.

5. **What type of bond is free from most taxes?**

- A. Corporate bonds
- B. Municipal bonds
- C. Government bonds
- D. Zero-coupon bonds

Answer: C. Government bonds

6. **Which term describes the extra yield demanded by investors for bonds with varying degrees of liquidity?**

- A. Taxability premium
- B. Default risk premium
- C. Liquidity premium
- D. Inflation premium

Answer: C. Liquidity premium

7. **What is the value of a stock with constant dividends, known as zero growth stocks, if the dividend is \$10 and the required rate of return is 20%?**

- A. \$20
- B. \$50
- C. \$200
- D. \$100

Answer: B. \$50

8. **If an investor requires a 25% return on a stock expected to pay a \$10 dividend and be worth \$70 in one year, what is the maximum price the investor would pay for the stock today?**

- A. \$80
- B. \$70
- C. \$64
- D. \$50

Answer: C. \$64

9. **What is the term for the difference in yields on treasury notes and bonds of different maturities when plotted?**

- A. Interest rate curve
- B. Discount curve
- C. Yield curve
- D. Bond curve

Answer: C. Yield curve

10. What type of premium compensates investors for the risk of default on a bond?

- A. Taxability premium
- B. Default risk premium
- C. Liquidity premium
- D. Interest rate risk premium

Answer: B. Default risk premium

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Zero Growth Stocks

1. What is the value of a zero growth stock with a \$10 dividend per year and a required return of 20%?

- A. \$50
- B. \$100
- C. \$20
- D. \$200

Answer: A. \$50

2. How is the value of a zero growth stock calculated?

- A. $P_0 = D \times RD \times RD \times R$
- B. $P_0 = D / RD / RD / R$
- C. $P_0 = R / DR / DR / D$
- D. $P_0 = D \times (1+R)D \times (1 + R)D \times (1+R)$

Answer: B. $P_0 = D / RD / RD / R$

Constant Growth Stocks

3. What will be the dividend in 5 years for a company that just paid a \$3 dividend and grows at 8% annually?

- A. \$4.00
- B. \$4.41
- C. \$3.24
- D. \$5.00

Answer: B. \$4.41

4. **How do you calculate the future dividend for a constant growth stock?**

- A. $D_t = D_0 \times (1+g)$ $D_t = D_0 \times (1 + g)$
- B. $D_t = D_0 \times g$ $D_t = D_0 \times g$
- C. $D_t = D_0 / (1+g)$ $D_t = D_0 / (1 + g)$
- D. $D_t = D_0 / g$ $D_t = D_0 / g$

Answer: A. $D_t = D_0 \times (1+g)$ $D_t = D_0 \times (1 + g)$

5. **What is the price of a stock with a \$2.30 dividend, 13% required return, and 5% growth rate?**

- A. \$25
- B. \$28.75
- C. \$30
- D. \$36.88

Answer: D. \$36.88

6. **What type of model is used to value a constant growth stock?**

- A. Zero Growth Model
- B. Dividend Growth Model
- C. Yield Model
- D. Interest Rate Model

Answer: B. Dividend Growth Model

7. **If a company's next dividend is \$4 and grows by 6% annually with a required return of 16%, what is the stock value today?**

- A. \$40
- B. \$45
- C. \$50
- D. \$60

Answer: A. \$40

8. **What is the dividend in four years for a company with a \$4 next dividend and a 6% growth rate?**

- A. \$4.24
- B. \$4.48
- C. \$4.76
- D. \$5.00

Answer: C. \$4.76

9. **What happens to the stock price if the growth rate exceeds the discount rate?**

- A. It decreases indefinitely.
- B. It becomes infinitely large.

- C. It remains constant.
- D. It equals the dividend.

Answer: B. It becomes infinitely large.

10. What does the dividend growth model implicitly assume about the stock price?

- A. It will decrease over time.
- B. It will grow at the same rate as the dividend.
- C. It remains unchanged.
- D. It will decrease at the same rate as the dividend.

Answer: B. It will grow at the same rate as the dividend.

Non-Constant Growth Stocks

11. What is the stock price today for a company paying a \$0.50 dividend in 5 years, growing at 10% annually, with a 20% required return?

- A. \$3.00
- B. \$4.50
- C. \$2.41
- D. \$5.00

Answer: C. \$2.41

12. What is the stock price three years from now if the expected dividends are \$1, \$2, and \$2.50, and the dividend grows at 5% thereafter with a 10% required return?

- A. \$20.00
- B. \$30.00
- C. \$50.00
- D. \$25.00

Answer: D. \$25.00

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Non-Constant Growth Stocks

1. CR Inc. has been growing at a rate of 30% per year. How long is this growth rate expected to last before it drops to 10% per year?

- A) 1 year
- B) 2 years
- C) 3 years
- D) 4 years
- **Correct Answer: C) 3 years**

2. If CR Inc. just paid a total dividend of \$5 million, what will be the dividend at the end of year 2, assuming a 30% growth rate?

- A) \$6.50 million
- B) \$8.45 million
- C) \$10.985 million
- D) \$11.985 million
- **Correct Answer: B) \$8.45 million**

3. **What is the formula to calculate the price of the stock at the end of the supernormal growth period?**

$$P_3 = \frac{D_3 \times (1 + g)}{R - g}$$

4. **If the required return is 20% and the long-run growth rate is 10%, what is the value of the stock at the end of year 3, given that D3 is \$10.985 million?**
- A) \$98.835 million
 - B) \$109.835 million
 - C) \$119.835 million
 - D) \$120.835 million
 - **Correct Answer: D) \$120.835 million**
5. **What is the formula to determine the present value of a stock with non-constant growth?**

$$P_0 = \frac{D_1}{(1 + R)} + \frac{D_2}{(1 + R)^2} + \dots + \frac{P_n}{(1 + R)^n}$$

6. **If the dividends are expected to grow at a constant rate indefinitely after a certain period, what model can be used to value the stock?**
- A) Dividend Discount Model
 - B) Dividend Growth Model
 - C) Price-Earnings Ratio Model
 - D) Gordon Growth Model
 - **Correct Answer: D) Gordon Growth Model**

Components of Required Return

7. **What are the two components of the total return RRR as per the dividend growth model?**
- A) Dividend yield and capital gains yield
 - B) Current yield and interest yield
 - C) Price-earnings ratio and growth rate
 - D) Dividend payout ratio and capital structure
 - **Correct Answer: A) Dividend yield and capital gains yield**
8. **If a stock is selling for \$20 per share, the next dividend is \$1 per share, and it is expected to grow by 10% indefinitely, what is the expected return?**
- A) 10%
 - B) 15%
 - C) 20%

- D) 25%
- **Correct Answer: B) 15%**

Stock Valuation Summary

9. According to the general case of stock valuation, the price today of a share of stock P0P0P0 is the present value of:
 - A) All future dividends
 - B) All future earnings
 - C) All future sales
 - D) All future growth rates
 - **Correct Answer: A) All future dividends**
10. In the constant growth rate model, if the dividend grows at a steady rate ggg, the price P0P0P0 can be calculated as:
 - A) $P0 = D0 / (R - g)$
 - B) $P0 = D1 / (R - g)$
 - C) $P0 = D2 / (R - g)$
 - D) $P0 = D3 / (R - g)$
 - **Correct Answer: B) $P0 = D1 / (R - g)$**

Common Stock Features

1. What is the primary method by which shareholders control a corporation?
 - A) Allocating dividends
 - B) Electing the board of directors
 - C) Determining executive compensation
 - D) Approving mergers and acquisitions
 - **Correct Answer: B) Electing the board of directors**
2. Which voting method allows shareholders to distribute their votes among candidates in a flexible manner?
 - A) Cumulative voting
 - B) Straight voting
 - C) Proportional voting
 - D) Consensus voting
 - **Correct Answer: A) Cumulative voting**
3. What mechanism ensures that shareholders can maintain their proportionate ownership in a company when new stock is issued?
 - A) Cumulative voting
 - B) Proportional voting
 - C) Preemptive right
 - D) Staggering
 - **Correct Answer: C) Preemptive right**

Preferred Stock Features

4. What characteristic differentiates preferred stock from common stock?

- A) Voting rights
 - B) Dividend priority
 - C) Convertibility
 - D) Liquidity
 - **Correct Answer: B) Dividend priority**
5. **Which term refers to the right of preferred stockholders to receive any unpaid dividends before common stockholders receive any dividends?**
- A) Liquidation preference
 - B) Cumulative dividend
 - C) Preferred claim
 - D) Seniority preference
 - **Correct Answer: B) Cumulative dividend**

Stock Market

6. **What is the function of a broker in the stock market?**
- A) Buying and selling securities for their own account
 - B) Offering stocks at initial public offerings (IPOs)
 - C) Matching buyers and sellers of securities
 - D) Setting bid and ask prices
 - **Correct Answer: C) Matching buyers and sellers of securities**
7. **Which market involves the trading of previously issued securities among investors?**
- A) Primary market
 - B) Initial public offering (IPO)
 - C) Secondary market
 - D) Over-the-counter market (OTC)
 - **Correct Answer: C) Secondary market**

Dividends and Stock Valuation

8. **What determines whether a dividend is paid by a corporation?**
- A) Shareholder vote
 - B) CEO decision
 - C) Board of directors' discretion
 - D) Government regulation
 - **Correct Answer: C) Board of directors' discretion**
9. **How are dividends typically treated for tax purposes in many jurisdictions?**
- A) Deductible business expense
 - B) Non-taxable income
 - C) Tax-deductible at corporate level
 - D) Taxable as ordinary income to shareholders
 - **Correct Answer: D) Taxable as ordinary income to shareholders**

Miscellaneous

10. Which model is used to estimate the intrinsic value of a stock based on expected future dividends?

- A) P/E Ratio Model
- B) Dividend Discount Model (DDM)
- C) Price-to-Book Model
- D) Return on Equity Model
- **Correct Answer: B) Dividend Discount Model (DDM)**

Preferred Stocks

1. What is the primary characteristic that differentiates preferred stock from common stock?

- A) Convertibility into common stock
- B) Voting rights in corporate governance
- C) Fixed dividend priority
- D) Liquidity preference
- **Correct Answer: C) Fixed dividend priority**

2. In the context of preferred stocks, what does cumulative dividend mean?

- A) Dividends are paid out of accumulated profits only
- B) Dividends accrue and must be paid before common stock dividends if skipped
- C) Dividends are paid at a fixed rate annually
- D) Dividends are paid in shares instead of cash
- **Correct Answer: B) Dividends accrue and must be paid before common stock dividends if skipped**

3. If a company issues cumulative preferred stocks and skips paying dividends in a particular year, what happens to these dividends?

- A) They are forfeited by the shareholders
- B) They are paid out of future profits before any common stock dividends are paid
- C) They are converted into common stock
- D) They are doubled the following year
- **Correct Answer: B) They are paid out of future profits before any common stock dividends are paid**

Stock Market

4. What is the primary function of a dealer in the stock market?

- A) Facilitating IPOs for new companies
- B) Matching buyers and sellers of previously issued securities
- C) Setting the bid and ask prices for stocks
- D) Voting on corporate governance issues
- **Correct Answer: B) Matching buyers and sellers of previously issued securities**

5. Which market involves the initial sale of new securities to investors?

- A) Primary market
- B) Secondary market
- C) Over-the-counter market
- D) Auction market
- **Correct Answer: A) Primary market**

6. What is the term used to describe the difference between the bid price and the ask price of a security?
- A) Strike price
 - B) Spread
 - C) Yield
 - D) Volatility
 - **Correct Answer: B) Spread**

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Preferred Stock Features

1. What is the primary characteristic of preferred stock that distinguishes it from common stock?
 - A) Convertibility into bonds
 - B) Fixed dividend priority
 - C) Voting rights in corporate governance
 - D) Option to receive dividends as shares
 - **Correct Answer: B) Fixed dividend priority**
2. Under what circumstances can directors decide not to pay dividends on preferred stock?
 - A) When the company has significant profits
 - B) Regardless of the company's net income
 - C) Only if common stockholders agree
 - D) If preferred shareholders vote against dividends
 - **Correct Answer: B) Regardless of the company's net income**
3. What happens to unpaid dividends on cumulative preferred stock?
 - A) They are forfeited by the shareholders
 - B) They are carried forward as arrearage
 - C) They are converted into common stock
 - D) They are doubled in the following year
 - **Correct Answer: B) They are carried forward as arrearage**

Capital Budgeting and NPV

4. What does Net Present Value (NPV) represent in capital budgeting?
 - A) Future cash flows discounted at the cost of equity
 - B) Value added to shareholders by an investment
 - C) Total profits expected from an investment
 - D) Capital raised from new securities
 - **Correct Answer: B) Value added to shareholders by an investment**
5. How is NPV calculated in discounted cash flow (DCF) valuation?
 - A) By dividing future cash flows by current market value
 - B) By subtracting initial investment from future cash flows
 - C) By applying a discount rate to future cash flows

- D) By adding up all future cash flows
 - **Correct Answer: C) By applying a discount rate to future cash flows**
6. According to the NPV rule, when should an investment be accepted?
- A) When the NPV is negative
 - B) When the NPV is zero
 - C) When the NPV is positive
 - D) When the NPV is undefined
 - **Correct Answer: C) When the NPV is positive**

Preferred Stock Features

1. Which statement about preferred stock is true?
 - A) Preferred shareholders have voting rights in corporate decisions.
 - B) Preferred dividends must always be paid before common dividends.
 - C) Preferred stockholders receive higher dividends than common stockholders.
 - D) Preferred stock can be converted into bonds at any time.
 - **Correct Answer: B) Preferred dividends must always be paid before common dividends.**
2. What is the primary characteristic of cumulative preferred stock?
 - A) Dividends can be skipped by directors.
 - B) Dividends are paid only if common shareholders approve.
 - C) Unpaid dividends accumulate and must be paid in future years.
 - D) Dividends are paid at a variable rate based on company profits.
 - **Correct Answer: C) Unpaid dividends accumulate and must be paid in future years.**

Capital Budgeting and NPV

3. What is the Net Present Value (NPV) of an investment project that costs \$50,000 and has a discounted cash flow of \$60,000? The discount rate is 10%.
 - A) \$10,000
 - B) \$6,000
 - C) \$5,000
 - D) \$4,000
 - **Correct Answer: A) \$10,000**
4. In capital budgeting, the discount rate used to calculate NPV is typically based on:
 - A) The company's net profit margin.
 - B) The average inflation rate over the past 10 years.
 - C) The risk associated with the investment project.
 - D) The current stock price of the company.
 - **Correct Answer: C) The risk associated with the investment project.**
5. Which statement best describes the NPV rule for investment decisions?
 - A) An investment should be accepted if the NPV is positive.
 - B) An investment should be rejected if the NPV is zero.
 - C) NPV is irrelevant in capital budgeting decisions.
 - D) NPV is only used for short-term projects.
 - **Correct Answer: A) An investment should be accepted if the NPV is positive.**

